

Vector Spaces and Subspaces

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Q1

- (i) The set S_1 is a subspace. It contains the zero vector, and if $u = (x_1, y_1, z_1)$ and $v = (x_2, y_2, z_2)$ belong to S_1 , then for any $c, d \in \mathbb{R}$,

$$c(x_1 + y_1 + z_1) + d(x_2 + y_2 + z_2) = 0.$$

Thus $cu + dv \in S_1$.

The set S_2 is not a subspace because it does not contain the zero vector.

The set S_3 is not a subspace. For example,

$$u = (1, 0, 1), \quad v = (0, 1, 1)$$

both belong to S_3 , but $u + v = (1, 1, 2)$ does not.

The set S_4 is a subspace by the definition of span. Any linear combination of vectors in S_4 is again a linear combination of $(1, 4, 0)$ and $(2, 2, 2)$.

The set S_5 is not a subspace. The vector $(0, 1, 2)$ belongs to S_5 , but its negative $(0, -1, -2)$ does not satisfy $0 \leq -1 \leq -2$.

- (ii) Because S and T are subspaces, both contain the zero vector. Hence $0 \in S \cap T$. Let $u, v \in S \cap T$ and let $c, d \in \mathbb{R}$. Since $u, v \in S$ and S is a subspace,

$$cu + dv \in S.$$

The same argument gives $cu + dv \in T$. Therefore $cu + dv \in S \cap T$, and $S \cap T$ is a subspace of V .

Q2

- (i) Every vector in the column space of $[A \ b]$ has the form

$$Ax + tb, \quad x \in \mathbb{R}^n, \quad t \in \mathbb{R}.$$

Suppose first that $b \in \mathcal{C}(A)$. Then $b = Ax_0$ for some $x_0 \in \mathbb{R}^n$, and

$$Ax + tb = A(x + tx_0) \in \mathcal{C}(A).$$

Thus $\mathcal{C}([A \ b]) \subseteq \mathcal{C}(A)$. The reverse inclusion follows by taking $t = 0$, so the two column spaces are equal.

Conversely, suppose $\mathcal{C}([A \ b]) = \mathcal{C}(A)$. The appended column b belongs to $\mathcal{C}([A \ b])$, and therefore $b \in \mathcal{C}(A)$. Finally, by the definition of the column space,

$$b \in \mathcal{C}(A) \iff b = Ax \text{ for some } x \in \mathbb{R}^n.$$

This is exactly the solvability of $Ax = b$.

(ii) Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Its column space is the x -axis in $\mathbb{R}^2$. Appending

$$b = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

enlarges the column space to all of $\mathbb{R}^2$. By contrast, appending

$$c = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

does not enlarge it because c already belongs to $\mathcal{C}(A)$.

Q3

(i) With

$$A = \begin{bmatrix} 1 & -3 & -1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix},$$

the equation is $A\mathbf{x} = 0$. The variable x is a pivot variable, while y and z are free. Solving for the pivot variable gives

$$x = 3y + z.$$

Therefore

$$\mathbf{x} = y \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

A basis for the nullspace is

$$\left\{ \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

(ii) A convenient particular solution is

$$\mathbf{x}_p = \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix}.$$

The complete solution is

$$\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n = \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad s, t \in \mathbb{R}.$$

This solution set is not a subspace because it does not contain the zero vector. It is an affine plane obtained by translating $\mathcal{N}(A)$ by $\mathbf{x}_p$.

Q4

- (i) Elimination on the augmented matrix begins with

$$\begin{bmatrix} 1 & 2 & -2 & b_1 \\ 2 & 5 & -4 & b_2 \\ 4 & 9 & -8 & b_3 \end{bmatrix}.$$

Apply $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - 4R_1$:

$$\begin{bmatrix} 1 & 2 & -2 & b_1 \\ 0 & 1 & 0 & b_2 - 2b_1 \\ 0 & 1 & 0 & b_3 - 4b_1 \end{bmatrix}.$$

Then $R_3 \leftarrow R_3 - R_2$ gives

$$\begin{bmatrix} 1 & 2 & -2 & b_1 \\ 0 & 1 & 0 & b_2 - 2b_1 \\ 0 & 0 & 0 & b_3 - b_2 - 2b_1 \end{bmatrix}.$$

The system is solvable if and only if

$$b_3 - b_2 - 2b_1 = 0.$$

- (ii) Assume that $b_3 = b_2 + 2b_1$. The second equation gives

$$y = b_2 - 2b_1.$$

Choose the free variable $z = t$. The first equation then gives

$$x = 5b_1 - 2b_2 + 2t.$$

Hence the complete solution is

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5b_1 - 2b_2 \\ b_2 - 2b_1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \quad t \in \mathbb{R}.$$

The first vector is one particular solution, and the second term contains all homogeneous solutions.

- (iii) The echelon form has two pivots, so the coefficient matrix has rank 2. Its nullspace is one-dimensional, with basis

$$\left\{ \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Q5

Place u_1, u_2, u_3 into the columns of a matrix:

$$U = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{bmatrix}.$$

Apply $R_2 \leftarrow R_2 - 3R_1$ and $R_3 \leftarrow R_3 - 2R_1$, then exchange rows 2 and 3:

$$U \longrightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -7 \\ 0 & -1 & -5 \end{bmatrix}.$$

Finally, $R_3 \leftarrow R_3 - 5R_2$ gives

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -5 \\ 0 & 0 & 18 \end{bmatrix}.$$

There are three pivots, so the only solution of

$$c_1u_1 + c_2u_2 + c_3u_3 = 0$$

is $c_1 = c_2 = c_3 = 0$. Thus u_1, u_2, u_3 are linearly independent. Three independent vectors in $\mathbb{R}^3$ form a basis for $\mathbb{R}^3$.

For the second triple,

$$v_1 + v_2 + v_3 = (0, 0, 0).$$

This is a nontrivial linear relation, so v_1, v_2, v_3 are dependent and do not form a basis for $\mathbb{R}^3$.

Q6

(i) Every polynomial in $\mathcal{P}_3$ has a unique representation

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3.$$

Therefore

$$\{1, x, x^2, x^3\}$$

is a basis for $\mathcal{P}_3$, and $\dim \mathcal{P}_3 = 4$.

(ii) The zero polynomial belongs to S . If $p, q \in S$ and $c, d \in \mathbb{R}$, then

$$(cp + dq)(1) = cp(1) + dq(1) = 0.$$

Thus $cp + dq \in S$, and S is a subspace.

The condition $p(1) = 0$ gives

$$a_0 + a_1 + a_2 + a_3 = 0,$$

so $a_0 = -a_1 - a_2 - a_3$. Consequently,

$$p(x) = a_1(x - 1) + a_2(x^2 - 1) + a_3(x^3 - 1).$$

The three polynomials $x - 1$, $x^2 - 1$, and $x^3 - 1$ are linearly independent because their leading nonzero terms have different degrees. Hence

$$\{x - 1, x^2 - 1, x^3 - 1\}$$

is a basis for S , and $\dim S = 3$.

Q7

- (i) The second row of A is twice the first, and every column is a multiple of $(1, 2)$. Therefore $\text{rank}(A) = 1$. A basis for the column space is

$$\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}, \quad \dim \mathcal{C}(A) = 1,$$

and a basis for the row space is

$$\{(1, 2, 4)\}, \quad \dim \mathcal{R}(A) = 1.$$

The equation $Ax = 0$ reduces to

$$x_1 + 2x_2 + 4x_3 = 0.$$

Thus

$$x = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -4 \\ 0 \\ 1 \end{bmatrix},$$

so a basis for the nullspace is

$$\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ 0 \\ 1 \end{bmatrix} \right\}, \quad \dim \mathcal{N}(A) = 2.$$

Finally, $A^T y = 0$ reduces to $y_1 + 2y_2 = 0$. A basis for the left nullspace is

$$\left\{ \begin{bmatrix} -2 \\ 1 \end{bmatrix} \right\}, \quad \dim \mathcal{N}(A^T) = 1.$$

- (ii) The first two columns of B are independent because

$$c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 5 \end{bmatrix} = 0$$

implies $c_1 + 2c_2 = 0$ and $2c_1 + 5c_2 = 0$, hence $c_1 = c_2 = 0$. Therefore $\text{rank}(B) = 2$. A basis for the column space is

$$\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \end{bmatrix} \right\}, \quad \dim \mathcal{C}(B) = 2.$$

The two rows are also independent, so a basis for the row space is

$$\{(1, 2, 4), (2, 5, 8)\}, \quad \dim \mathcal{R}(B) = 2.$$

Solving $Bx = 0$ gives

$$x_2 = 0, \quad x_1 = -4x_3.$$

Therefore a basis for the nullspace is

$$\left\{ \begin{bmatrix} -4 \\ 0 \\ 1 \end{bmatrix} \right\}, \quad \dim \mathcal{N}(B) = 1.$$

Since the columns span $\mathbb{R}^2$, the left nullspace contains only the zero vector. Its basis is empty and

$$\dim \mathcal{N}(B^T) = 0.$$

(iii) For A , the dimensions satisfy

$$\dim \mathcal{R}(A) + \dim \mathcal{N}(A) = 1 + 2 = 3$$

and

$$\dim \mathcal{C}(A) + \dim \mathcal{N}(A^T) = 1 + 1 = 2.$$

The row-space basis vector is orthogonal to both nullspace basis vectors:

$$(1, 2, 4) \cdot (-2, 1, 0) = 0, \quad (1, 2, 4) \cdot (-4, 0, 1) = 0.$$

The column-space and left-nullspace basis vectors are also orthogonal:

$$(1, 2) \cdot (-2, 1) = 0.$$

For B , the dimensions satisfy

$$\dim \mathcal{R}(B) + \dim \mathcal{N}(B) = 2 + 1 = 3$$

and

$$\dim \mathcal{C}(B) + \dim \mathcal{N}(B^T) = 2 + 0 = 2.$$

The nullspace basis vector is orthogonal to both rows:

$$(1, 2, 4) \cdot (-4, 0, 1) = 0, \quad (2, 5, 8) \cdot (-4, 0, 1) = 0.$$

The left nullspace is the zero space, which is orthogonal to every vector in $\mathcal{C}(B)$.

Q8

(i) Write the columns of B as $b_1, \dots, b_p$. Then

$$AB = [Ab_1 \quad \dots \quad Ab_p].$$

Every column Ab_j is a linear combination of the columns of A . Therefore

$$\mathcal{C}(AB) \subseteq \mathcal{C}(A).$$

Taking dimensions gives

$$\text{rank}(AB) \leq \text{rank}(A).$$

(ii) If $x \in \mathcal{N}(B)$, then $Bx = 0$, and consequently

$$ABx = A0 = 0.$$

Thus $x \in \mathcal{N}(AB)$, proving that

$$\mathcal{N}(B) \subseteq \mathcal{N}(AB).$$

Both nullspaces are subspaces of $\mathbb{R}^p$, so

$$\dim \mathcal{N}(B) \leq \dim \mathcal{N}(AB).$$

By rank-nullity,

$$p - \text{rank}(B) \leq p - \text{rank}(AB),$$

and hence

$$\text{rank}(AB) \leq \text{rank}(B).$$

(iii) For every $x \in \mathbb{R}^n$,

$$(C + D)x = Cx + Dx.$$

Therefore every vector in $\mathcal{C}(C + D)$ belongs to the sum of subspaces

$$\mathcal{C}(C) + \mathcal{C}(D) = \{u + v : u \in \mathcal{C}(C), v \in \mathcal{C}(D)\}.$$

If bases for $\mathcal{C}(C)$ and $\mathcal{C}(D)$ contain r_C and r_D vectors, their union spans $\mathcal{C}(C) + \mathcal{C}(D)$. Consequently,

$$\begin{aligned} \text{rank}(C + D) &= \dim \mathcal{C}(C + D) \\ &\leq \dim(\mathcal{C}(C) + \mathcal{C}(D)) \\ &\leq r_C + r_D \\ &= \text{rank}(C) + \text{rank}(D). \end{aligned}$$

(iv) Take

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Both A and B have rank 1, while $AB = 0$ has rank 0. Hence

$$\text{rank}(AB) = 0 < \text{rank}(A) = 1$$

and

$$\text{rank}(AB) = 0 < \text{rank}(B) = 1.$$